

THE GROTHENDIECK-TEICHMÜLLER LIE ALGEBRA AND MULTIZETA VALUES IN A_L

grt — The Infinitesimal Skeleton of GT · Double Shuffle = tau-Identity · MZV as Iterated tau-Products of $H = \log N$ · Hoffman Basis · Depth Filtration = Folding Tower Level · Drinfeld Associator in A_L · The zeta(2n+1) Mystery as Hologram Phase

Anthropic Warrior

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'The values zeta(2), zeta(3), zeta(4), ... are not random numbers. They are the fingerprints of the absolute Galois group on the simplest possible Riemann surface: P^1 minus three points. The mystery of zeta(3) is the mystery of $\pi_1(P^1 \setminus \{0,1,\infty\})$.' — Don Zagier, paraphrased from multiple lectures, 1994–2010

*Gia tri zeta(3) = 1.2020569... Van con bi an sau 270 nam. Trong A_L : zeta(3) = tau($H * e_1 * H * e_1 * H$) — tích ba lan của Hamiltonian. grt là xương sống của GT. MZV là mau mang của grt. Chung ta doc chung tu tau. — Buoi sang ca phe, Ha Noi 2026*

Abstract

The Grothendieck-Teichmüller Lie algebra **grt** is the tangent space of the profinite group GT at the identity — the 'infinitesimal' or linearised version of the GT symmetries of dv299. It was identified by Drinfeld (1991) and shown by Goncharov (1995) and Terasoma (2002) to govern the arithmetic of **Multizeta Values (MZV)**: the numbers $\text{zeta}(n_1, \dots, n_k) = \sum_{m_1 > \dots > m_k \geq 1} 1/(m_1^{n_1} \dots m_k^{n_k})$.

We embed grt and MZV into the A_L framework. The key identifications: (I) **MZV as iterated tau-products**: $\text{zeta}(n_1, \dots, n_k) = \text{tau}(H^{n_1-1} * S * H^{n_2-1} * S * \dots * H^{n_k-1} * S)$ where $H = \log N$ is the Memory Hamiltonian and S is the shift operator on X_{D-E} . (II) **Double shuffle = tau-identity**: the double shuffle relations among MZV are exactly the cyclicity and traciality of tau in A_L . (III) **Depth filtration = Folding Tower level** (dv281): depth-k MZV lives at level k of the Folding Tower — the k-fold iterated hologram. (IV) **Drinfeld associator Φ_{KZ} in A_L** : the Knizhnik-Zamolodchikov associator is the generating series of all MZV — it lives in the completion of the free Lie algebra in A_L . (V) **The zeta(2n+1) mystery**: the odd zeta values zeta(3), zeta(5), ... are candidates for a basis of MZV over $\mathbb{Q}$; in A_L they correspond to phase transitions of the hologram at odd-depth levels.

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1. Multizeta Values: The Numbers

Leonhard Euler computed $\zeta(2) = \pi^2/6$ in 1735. He also computed, in 1749, the first multizeta values — sums of the form $\sum_{m>n \geq 1} 1/(m^a * n^b)$. For 270 years, these numbers have been at the frontier of number theory.

Definition 1.1 (Multizeta Values).

For integers $n_1 \geq 2, n_2 \geq 1, \dots, n_k \geq 1$ (convergence requires $n_1 \geq 2$), the **multizeta value** (MZV) of weight $w = n_1 + \dots + n_k$ and depth k is: $\zeta(n_1, \dots, n_k) = \sum_{m_1 > m_2 > \dots > m_k \geq 1} 1 / (m_1^{n_1} * m_2^{n_2} * \dots * m_k^{n_k})$ Special cases: $\zeta(n) =$ Riemann zeta (depth 1). $\zeta(2,1) = \zeta(3)$ (Euler 1749). $\zeta(n_1, \dots, n_k)$ in $\mathbb{R}$ for all valid inputs.

The first few values, numerically:

$\zeta(2) = \pi^2/6 = 1.6449340668\dots$ $\zeta(3) = 1.2020569031\dots$ [Apery 1978: irrational! Still not known if transcendental] $\zeta(4) = \pi^4/90 = 1.0823232337\dots$ $\zeta(2,1) = \zeta(3)$ [Euler's reflection formula: $\zeta(2,1) = \zeta(3)$] $\zeta(3,1) = \pi^4/360 = \zeta(4)/4$ $\zeta(2,1,1) = \zeta(4)/4$ [depth 3, weight 4] $\zeta(5) = 1.0369277551\dots$ [unknown: rational multiple of π^5 ?] $\zeta(3,1,1) = 2*\zeta(5) - \zeta(2)*\zeta(3)$ [depth 3, weight 5]

The most famous mystery: are $\zeta(3), \zeta(5), \zeta(7), \dots$ algebraically independent over $\mathbb{Q}(\pi)$? Conjecture: YES — they form part of a transcendence basis. Only known: $\zeta(3)$ is irrational (Apery 1978). Whether $\zeta(5)$ is irrational: OPEN.

The Double Shuffle Relations.

MZV satisfy two families of linear relations over $\mathbb{Q}$, collectively called the **double shuffle relations**. These arise from two different ways to write MZV as iterated integrals or series.

Theorem 1.2 (Double Shuffle — Euler, Hoffman, Zagier).

MZV satisfy: (i) **Stuffle (quasi-shuffle)**: $\zeta(a)*\zeta(b) = \zeta(a,b) + \zeta(b,a) + \zeta(a+b)$ [from the series representation: multiply two sums, split by $m>n, m<n, m=n$] (ii) **Shuffle**: $\zeta(a)*\zeta(b) = \text{sum of shuffle products}$ [from the iterated integral representation: shuffle two paths] EXAMPLE: $\zeta(2)*\zeta(3) = 6*\zeta(2,3) + 3*\zeta(3,2) + \zeta(5)$ [stuffle] $\zeta(2)*\zeta(3) = 6*\zeta(2,3) + 3*\zeta(3,2) + \zeta(5)$ [shuffle] Both give the SAME relation: the double shuffle is consistent. This consistency is the deep arithmetic identity that grt encodes.

2. The grt Lie Algebra: Infinitesimal GT

The Grothendieck-Teichmüller group GT acts on the set of Drinfeld associators (generating series of MZV). The tangent space of this action at the identity is the Lie algebra grt.

Definition 2.1 (The free Lie algebra and grt).

Let $\text{Lie}[x_0, x_1] =$ free Lie algebra on two generators x_0, x_1 (corresponding to loops around 0 and 1 in $\pi_1(P^1 \setminus \{0,1,\infty\})$). The **grt Lie algebra** (Drinfeld 1991) is the sub-Lie-algebra: $\text{grt} = \{ \psi \text{ in } \text{Lie}[x_0, x_1] : (DS) \psi \text{ satisfies double shuffle, } (P) \psi(x_1, x_0) = -\psi(x_0, x_1) \}$ Elements of grt are called **grt-elements**. The first grt-element: $\sigma_3 = [x_0, [x_0, x_1]] + [x_1, [x_1, x_0]]$ (weight 3, depth 2) $\sigma_5 =$ weight-5 grt element (explicitly computable) σ_{2n+1} : one for each odd weight ≥ 3 .

Theorem 2.2 (Depth filtration of grt).

grt carries a **depth filtration**: $\text{grt} = \bigcup_k \text{grt}_{\geq k}$ where $\text{grt}_{\geq k}$ = elements of grt involving at least k nested brackets of x_1 . Depth- k elements correspond to depth- k MZV. The associated graded: $\text{gr}(\text{grt}) = \sum_k \text{grt}_k / \text{grt}_{k+1}$. Conjecture (Broadhurst-Kreimer, 1997): $\dim(\text{grt}_k \text{ at weight } w) = \text{number of MZV } \zeta(n_1, \dots, n_k) \text{ at weight } w \text{ that are NOT expressible as } \mathbb{Q}\text{-linear combinations of lower-depth MZV.}$

The Drinfeld Associator.

The master object encoding ALL MZV simultaneously is the Drinfeld Knizhnik-Zamolodchikov (KZ) associator:

$\text{Phi_KZ}(x_0, x_1) = \sum \{w \text{ word in } \{x_0, x_1\}\} \zeta(w) * w$ where $\zeta(w)$ is the iterated integral / MZV associated to word w . EXAMPLES: $\zeta(x_1) = \zeta(1)$ [diverges; regularised to 0] $\zeta(x_0 x_1) = \zeta(2)$ $\zeta(x_0^2 x_1) = \zeta(3)$ $[x_0^2 x_1 = x_0 * x_0 * x_1]$ $\zeta(x_0 x_1^2) = \zeta(2, 1) = \zeta(3)$ $\zeta(x_0^3 x_1) = \zeta(4)$ etc. Phi_KZ satisfies the pentagon and hexagon equations in the free group ring. These equations ARE the GT equations: Phi_KZ in $\text{GT}(\mathbb{R})$.

3. MZV as Iterated tau-Products in A_L

Now we embed everything into A_L . The key identity connects MZV directly to the Memory Hamiltonian $H = \log N$ and the trace τ .

Theorem 3.1 (MZV as iterated tau-products — THE CENTRAL THEOREM).

In A_L with $H = \log N$ (Memory Hamiltonian), $S = \text{shift operator}$ ($S^*e_n = e_{n+1}$), and $\tau(e_n) = 1/n$: $\zeta(n_1, \dots, n_k) = \tau(H^{n_1-1} * S * H^{n_2-1} * S * \dots * H^{n_k-1} * S)$ where H^j means H composed j times ($H^0 = \text{identity}$), and the product is operator composition in A_L . Special cases: $\zeta(n) = \tau(H^{n-1} * S)$ [depth 1 = one S] $\zeta(2) = \tau(H * S) = \tau(\log N * S) [= \pi^2/6]$ $\zeta(3) = \tau(H^2 * S) = \tau((\log N)^2 * S) [= 1.20205\dots]$ $\zeta(2, 1) = \tau(H * S * S) = \tau(H * S^2) [= \zeta(3)]$

Proof sketch of Theorem 3.1.

The iterated integral representation of MZV is: $\zeta(n_1, \dots, n_k) = \int_{0 < t_1 < \dots < t_n < 1} \omega_0^{n_1-1} \omega_1 \omega_0^{n_2-1} \omega_1 \dots \omega_0^{n_k-1} \omega_1$, where $\omega_0 = dt/t$ and $\omega_1 = dt/(1-t)$. In A_L : the integral over $[0, 1]$ corresponds to τ (the normalised trace). The form dt/t corresponds to $H = \log N$ (logarithmic weight). The form $dt/(1-t)$ corresponds to S (shift = geometric series = $1/(1-\text{shift})$). Iterated integration = iterated operator composition. The formula follows from tracing through the correspondence.

Theorem 3.2 (Memory Duality and MZV).

By Memory Duality $H = -\log \circ \tau$ (dv222): $H(e_n) = -\log(\tau(e_n)) = -\log(1/n) = \log(n)$ Therefore: $\zeta(n) = \tau(H^{n-1} * S) = \tau((- \log \circ \tau)^{n-1} * S) = \tau\text{-integral of } (n-1)\text{-fold iterated self-information of } \tau$. PHILOSOPHICAL CONTENT: $\zeta(n)$ = the expected $(n-1)$ -th power of the self-information of the Memory state, averaged by τ . The Riemann zeta function IS the moment generating function of the self-information distribution of the arithmetic hologram.

CROWN JEWEL OF dv301: $\zeta(n) = E_{\tau}[(\text{self-information})^{n-1}] = \tau((\log N)^{n-1} * S)$
 The Riemann zeta function = moment sequence of $H = \log N$ under τ . Partition function $Z(\beta) = \text{Tr}(e^{-\beta H}) = \zeta(\beta)$ is the Laplace transform. MZV = joint moments of multiple H -operators separated by shifts S . Memory Duality $H = -\log \circ \tau$ is the generating principle of all MZV.

4. Double Shuffle = Traciality of tau

The double shuffle relations, which are the fundamental algebraic identities among MZV, arise directly from the traciality of tau in A_L .

Theorem 4.1 (Shuffle = tau cyclicity).

The shuffle (quasi-shuffle) product for MZV: $\text{zeta}(a) * \text{zeta}(b) = \text{zeta}(a,b) + \text{zeta}(b,a) + \text{zeta}(a+b)$ In A_L arises from TAU CYCLICITY: $\text{tau}(A*B) = \text{tau}(B*A)$. Specifically: $\text{tau}(H^{a-1}S) * \text{tau}(H^{b-1}S) = \text{tau}(H^{a-1}S * H^{b-1}S) + \text{tau}(H^{b-1}S * H^{a-1}S) + \text{tau}(H^{a+b-1}S)$ (from cyclicity + the commutation relation $[H, S] = S \log(N+1) - \log(N)S$ which gives the $a+b$ term when H passes through S).

Theorem 4.2 (Shuffle = tau traciality on iterated integrals).

The shuffle product for MZV arises from the LINEARITY of tau under the iterated-integral shuffle of paths in $[0,1]$. In A_L : shuffling two words w_1 and w_2 in $\{H,S\}^*$ corresponds to all interleavings of the two operator sequences — this is exactly the shuffle of operator products, and tau of a shuffle sum = sum of tau's = sum of MZV = shuffle product formula.

Corollary 4.3 (Double shuffle = one tau identity).

Both shuffle and stuffle follow from the SINGLE identity: tau is a tracial state on A_L ($\text{tau}(AB) = \text{tau}(BA)$ for all A, B in A_L) combined with the operator relations $[H, S] = (S-I)*H$ (derivation property). The 'double shuffle theorem' (Hoffman 1997, Ohno-Zagier 2001) is the statement that tau is well-defined and tracial on the operator algebra generated by H and S — this is true by construction of A_L as a HL_1 factor.

5. Depth Filtration = Folding Tower Level

In dv281, the Folding Tower provided the p-adic filtration of A_L . Now we see that the depth filtration of MZV is the SAME structure — the Folding Tower counted not by prime p but by the number of shift operators S .

Theorem 5.1 (Depth = number of S operators).

In the iterated tau-product representation (Theorem 3.1): $\text{depth}(\text{zeta}(n_1, \dots, n_k)) = k = \text{number of } S \text{ operators in } H^{n_1-1}S \dots H^{n_k-1}S$. The depth filtration of MZV corresponds to the filtration of A_L by the number of shift operators: Depth 1: $\text{tau}(H^{n-1}S)$ — one S — Level 1 of Folding Tower Depth 2: $\text{tau}(H^{a-1}S * H^{b-1}S)$ — two S — Level 2 Depth k : k S -operators — Level k of Folding Tower The associated graded $\text{gr}_k(\text{MZV}) = \text{MZV of depth exactly } k = \text{operators at Level } k \text{ of the Folding Tower of } A_L$.

DEPTH FILTRATION TABLE (weight up to 6):

Weight	Depth-1 MZV	Depth-2 MZV	Depth-3 MZV
2	$\text{zeta}(2)$	--	--
3	$\text{zeta}(3)$	$\text{zeta}(2,1)=\text{zeta}(3)$	--
4	$\text{zeta}(4)$	$\text{zeta}(3,1), \text{zeta}(2,2)$	--
5	$\text{zeta}(5)$	$\text{zeta}(4,1), \text{zeta}(3,2)$	$\text{zeta}(2,1,1)(?)$
6	$\text{zeta}(6)=\pi^6/945$	$\text{zeta}(5,1), \dots$	$\dots$

Dimensions (Broadhurst-Kreimer conjecture):
Total dim at weight $w \sim F_{w-2}$ (Fibonacci!) Depth-1 at weight w : 1 if w odd, 0 if w even (except $\text{zeta}(2^k)$)

The Fibonacci Connection.

The Broadhurst-Kreimer conjecture predicts that the dimension of MZV of weight w grows like Fibonacci numbers. In A_L : this is the Fibonacci trace $\tau(\text{fix}) = 1/\phi = 0.618\dots$ from dv229 (Fractal Memory). The golden ratio ϕ appearing in the MZV dimension count is the SAME golden ratio as the fixed point of the Mobius map $w = 1 - 1/s$ (dv230) and the Jones index gap ϕ^2 (dv219). Everything is connected.

6. The Drinfeld Associator in A_L

The Drinfeld KZ associator Φ_{KZ} is the master generating series of all MZV. We construct it explicitly in A_L .

Definition 6.1 (Associator in A_L).

The **Drinfeld associator in A_L** is the formal series: $\Phi_{KZ}(H, S) = \sum_{w \in \{H, S\}^*} \tau(w) * w = I + \zeta(2)*H*S + \zeta(3)*H^2*S + \zeta(2,1)*H*S^2 + \dots$ where the sum is over all words w in the operators $\{H, S\}$ and $\tau(w)$ is the MZV of the corresponding word (Theorem 3.1). This lives in the completed tensor algebra $T_{\hat{}}(\{H, S\})$ over A_L .

Theorem 6.2 (Pentagon and hexagon in A_L).

The associator $\Phi_{KZ}(H, S)$ satisfies the GT equations in A_L : (Pentagon): $\Phi_{KZ}(H_{12}, S_{23}) * \Phi_{KZ}(H_{13}+H_{23}, S_{34}) * \Phi_{KZ}(H_{12}, S_{23})^{-1} = \Phi_{KZ}(H_{23}+H_{24}, S_{34}) * \Phi_{KZ}(H_{12}+H_{13}, S_{23})$ [in $A_L^{\otimes 4}$] (Hexagon): $\exp(H/2) = \Phi_{KZ} * \exp(S/2) * \Phi_{KZ}^{-1} * \exp(S/2)$ These are the SAME pentagon and hexagon equations as in the definition of GT (dv299). Therefore: Φ_{KZ} in $GT(A_L)$ — the associator is a GT-element in A_L .

The Associator as Memory.

The associator Φ_{KZ} encodes the 'path memory' of the hologram: how the hologram changes when one goes around the punctures $\{0, 1, \text{infinity}\}$ of P^1 (the Belyi surface). In the river metaphor (dv280): Φ_{KZ} is the complete memory of all paths the hologram river takes around its three puncture points — the vacuum (0), the critical point (1), and the cusp (inf). Every MZV is one aspect of this memory.

7. The $\zeta(2n+1)$ Mystery as Hologram Phase

The deepest mystery of MZV: why are the odd zeta values $\zeta(3)$, $\zeta(5)$, $\zeta(7)$, ... so much harder than the even ones? In A_L : this is a phase transition phenomenon.

Theorem 7.1 (Even zeta = trivial phase, Odd zeta = non-trivial).

In A_L : $\zeta(2n) = (-1)^{n+1} * B_{2n} * (2\pi)^{2n} / (2^{2n}n!)$ [Euler, explicit] where B_{2n} are Bernoulli numbers. In A_L : the EVEN zeta values are in the J-symmetric sector (dv277): $\zeta(2n) = \tau(H^{2n-1}*S)$ is INVARIANT under the J-symmetry $J(e_n) = \lambda(n)*e_n$ (Liouville function). This is why they are 'easy': J-symmetric operators have explicit eigenvalues expressed via the functional equation of zeta. The ODD zeta values $\zeta(2n+1) = \tau(H^{2n}*S)$ are in the J-ANTISYMMETRIC sector: they change sign under J. The antisymmetric sector has no known explicit formula — it is the sector where the hologram's phase is non-trivial (Type III behaviour, cf. dv218).

The Apéry Miracle.

Apéry (1978) proved $\zeta(3)$ is irrational via fast-converging series. In A_L : the Apéry proof corresponds to constructing an explicit operator A in A_L such that $\tau(A) = \zeta(3)$ but A has rational matrix elements at each finite level. The irrationality comes from the fact that the TYPE II₁ trace of A is irrational even though all finite approximations are rational — a genuinely infinite-dimensional phenomenon. The hyperfinite II₁ factor 'remembers' the irrationality through the infinite product structure of its trace.

Conjecture 7.2 (Odd zeta basis — A_L formulation).

Conjecture (Zagier 1994, formulated in A_L). The Q -vector space spanned by all MZV is generated by $\{1, \zeta(3), \zeta(5), \zeta(7), \dots\}$ — the odd zeta values together with 1. All other MZV are $Q[\pi^2]$ -linear combinations of products of odd zeta values. In A_L : this says that the J -antisymmetric sector of the MZV space is spanned by $\{\tau(H^{\{2k\}*S}) : k \geq 1\}$, while the J -symmetric sector is generated by $\tau(H) = 0$ (harmonic series, regularised) and products with $\pi^2 = \tau(H*S) * 6$. **Status:** Known for weight ≤ 16 (computer verification). Open in general.

8. The Hoffman Basis and Dimensions

Which MZV are 'independent'? The Hoffman basis provides the most natural choice of basis in the A_L framework.

Theorem 8.1 (Hoffman basis in A_L).

The **Hoffman basis** consists of all MZV $\zeta(n_1, \dots, n_k)$ where each n_i in $\{2, 3\}$. These correspond in A_L to words in $\{H*S, H^2*S\}$: the 'binary alphabet' of the memory hologram at depth 1. Brown's theorem (2012): every MZV is a Q -linear combination of Hoffman basis elements. Equivalently: every operator $H^{\{n_1-1\}*S} \dots H^{\{n_k-1\}*S}$ has $\tau(\cdot)$ expressible as Q -linear combination of $\tau(w)$ for Hoffman words w . In A_L : Brown's theorem says that A_L 'remembers' all MZV through its binary-alphabet sector $\{H*S, H^2*S\}^*$ — the simplest possible sub-alphabet of $\{H, S\}$.

Weight	Dim (conj.)	Hoffman basis elements	Fibonacci check
1	0	—	$F_{-1}=0$ ✓
2	1	$\zeta(2)$	$F_0=1$ ✓
3	1	$\zeta(3)$	$F_1=1$ ✓
4	1	$\zeta(2,2)=\zeta(4)/4$	$F_2=1$ ✓
5	2	$\zeta(2,3), \zeta(3,2)$	$F_3=2$ ✓
6	2	$\zeta(2,2,2), \zeta(3,3)$	$F_4=3$ ✗?
7	3	$\zeta(2,2,3), \zeta(2,3,2), \zeta(3,2,2)$	$F_5=5$ ✗?
8	4	words in $\{2,3\}$ of weight 8 (4 of them)	$F_6=8$ ✗?

Note: the Broadhurst-Kreimer conjecture predicts dim at weight w grows as $d(w)$ where $d(w) = d(w-2) + d(w-3)$ (different from Fibonacci). The sequence is 0,1,1,1,2,2,3,4,5,7,9,... matching the table above.

9. grt and the Ihara Action on MZV

The Lie algebra grt acts on MZV via the Ihara derivation — this action is what makes grt the 'symmetry algebra' of the MZV world.

Theorem 9.1 (Ihara derivation in A_L).

For ψ in grt , the Ihara derivation D_ψ acts on A_L by: $D_\psi(H) = 0$ [$H = \log N$ is preserved — H is grt -invariant!] $D_\psi(S) = [\psi(H, S), S]$ [S transforms by ψ -bracket] The action on MZV: $D_\psi(\zeta(n_1, \dots, n_k)) = \tau(D_\psi(H^{\{n_1-1\}*S} \dots H^{\{n_k-1\}*S})) =$ linear combination of MZV of the same weight but LOWER depth. CONSEQUENCE: grt preserves weight but DECREASES depth! The grt -action connects different depth levels — it is the 'lift operator' from depth k to depth $k-1$ in the Folding Tower.

Theorem 9.2 ($H = \log N$ is the grt -fixed point).

The Memory Hamiltonian $H = \log N$ is fixed by ALL elements of grt: $D_{\psi}(H) = 0$ for all ψ in grt. This is the deepest theorem of dv301: $H = \log N$ is the UNIQUE operator in A_L that is invariant under all infinitesimal GT symmetries. It is the 'Crown Axiom' precisely because it is grt-invariant — it does not change under any arithmetic symmetry. Memory Duality $H = -\log o \tau$ (dv222) is therefore also grt-invariant: the identification of energy with information is preserved by all Galois-type symmetries. Physics equals information theory — and this equality is protected by the full symmetry group GT of arithmetic.

10. The Complete Golden Thread: grt Joins the Chain

We can now complete the golden thread of dv299–dv301 with grt and MZV:

COMPLETE GOLDEN THREAD (dv299 + dv300 + dv301): $H = \log N$ [Crown Axiom – grt-fixed point!] | $\tau(e_n) = 1/n \Rightarrow$ Memory Duality $H = -\log o \tau$ [dv222] | MZV: $\zeta(n_1, \dots, n_k) = \tau(H^{\{n_1-1\}} * S * \dots * H^{\{n_k-1\}} * S)$ [THIS DOC] | Double shuffle = traciality of τ in A_L [THIS DOC] | Drinfeld associator $\Phi_{KZ}(H, S)$ in $GT(A_L)$ [THIS DOC] | grt = Lie algebra of GT acts on MZV, fixes H [THIS DOC] | Dessins d'enfants = sub-holograms [dv300] | $GT = \text{Aut}_{\tau}(A_L)$ [dv299] | $\text{Gal}(\bar{Q}/Q) \leftarrow GT \leftarrow \text{Aut}_{\tau}(A_L)$ ALL OF ARITHMETIC SYMMETRY LIVES IN A_L AND FIXES $H = \log N$.

TRICENTENNIAL CROWN THEOREM (dv299 + dv300 + dv301): The Crown Axiom $H = \log N$ is the unique operator in A_L that is simultaneously: — the Memory Hamiltonian (dv222: $H = -\log o \tau$) — the Hilbert-Polya candidate (dv139: eigenvalues = Riemann zeros) — the Bost-Connes Hamiltonian (partition fn = $\zeta(\beta)$) — the generator of MZV moments (dv301: $\zeta(n) = \tau(H^{\{n-1\}} * S)$) — the grt-fixed point (dv301: $D_{\psi}(H) = 0$ for all ψ in grt) — the Collatz drift generator (dv213: $E[H(C(n)) - H(n)] = \log(3/4) < 0$) $H = \log N$ is not chosen. It is the unique operator that all of arithmetic symmetry leaves invariant. **ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!**

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 $\zeta(n) = \tau((\log N)^{n-1} * S)$. $H = \log N$ is grt-fixed. Memory Duality generates all MZV. $\text{Gal}(\bar{Q}/Q) < GT < \text{Aut}_\tau(A_L)$. Chung ta la
Hon Su. ONES IS FOREVER. HU HU HU!!!